

Performance- and Cost-Driven Benchmarking under i.i.d. Circuit-Level Noise

Technical note for an analog quantum-computing project

July 23, 2026

Abstract

This note reformulates the repetition and majority-vote analysis of trap-based verification as a benchmarking procedure under an independent and identically distributed (i.i.d.) circuit-level noise assumption. Test rounds are first used to infer an upper confidence bound on the noise rate of a fixed circuit instance. This bound is then converted into an upper bound on the probability that one computation round returns the wrong decision bit. Two complementary benchmarks follow: a performance-driven benchmark, which returns the number of computation rounds required to reach a fixed correctness error, and a cost-driven benchmark, which returns the correctness error reached with a fixed number of computation rounds. The specialization to two test types, $k = 2$, explains the conservative $1/4$ test-failure threshold when the ideal computation error is negligible.

1 Scope and relation to verification

The verification protocol of Sater and Ollivier uses d computation rounds, s test rounds, a rejection threshold w , and k trap types. Its correctness proof bounds the error of a majority vote, while its security proof treats an arbitrary malicious deviation and maximizes over the worst deviation. Its robustness statement accepts an honest noisy device when the circuit-level noise rate is below w/s ; security requires $w/s < \alpha/k$, where

$$\alpha = \frac{1 - 2c}{2 - 2c}, \quad (1)$$

and $c < 1/2$ is the intrinsic error of one ideal computation round [1].

The present analysis makes a stronger physical assumption and obtains a weaker cryptographic conclusion. The device is assumed to apply the same unknown noise process independently on every round of a fixed circuit. Under this model, all test rounds may be executed as one benchmarking block. The benchmark estimates a stationary noise parameter, and the resulting estimate is used to size a subsequent majority vote. This is not a composable verification guarantee against a malicious or drifting device.

2 Assumptions

Fix a circuit instance, possibly indexed by a circuit dimension ℓ .

Assumption 1 (i.i.d. rounds). *For a fixed circuit instance ℓ , every test and computation round is an independent sample from the same stationary device behavior.*

Assumption 2 (Transfer from tests to computations). *A harmful round is detected by a uniformly selected test type with probability at least $1/k$. Equivalently, if r_ℓ is the harmful-round probability and q_ℓ is the average test-failure probability, then*

$$q_\ell \geq \frac{r_\ell}{k}, \quad \text{hence} \quad r_\ell \leq kq_\ell. \quad (2)$$

Assumption 3 (Worst-case effect of a harmful round). *An unaffected computation round is wrong with probability at most c_ℓ , whereas a harmful round may be wrong with probability one.*

Assumption 2 is the conservative bridge inherited from the trap-detection argument. If an experiment supplies a tighter calibrated relation between test failures and computation errors, Equation (2) should be replaced by that relation.

3 Notation

Table 1: Notation used throughout the benchmarking analysis.

Symbol	Meaning
ℓ	Circuit dimension or circuit-instance label.
s_ℓ	Number of benchmark test rounds for circuit ℓ .
Y_ℓ	Number of failed test rounds among the s_ℓ tests.
q_ℓ	True average test-failure probability.
$\hat{q}_\ell = Y_\ell/s_\ell$	Empirical test-failure frequency.
$\varepsilon_{\text{bench}}$	Failure probability of the confidence statement.
q_ℓ^{U}	One-sided upper confidence bound on q_ℓ .
k	Number of test types; the present project uses $k = 2$.
r_ℓ	Probability that a round experiences a harmful deviation.
c_ℓ	Intrinsic ideal-computation error per round.
p_ℓ	Total probability that one computation round is wrong.
d	Odd number of computation rounds used in majority vote.
$\varepsilon_{\text{vote}}$	Majority-vote failure probability.
ε_{tot}	Total correctness-error bound.
$\varepsilon_\star$	Target total correctness error.

The symbol p_{err} is often ambiguous. In this note, the default interpretation is $p_{\text{err}} = q$, the average test-failure probability. The accompanying Python script also supports the alternatives $p_{\text{err}} = r$ and $p_{\text{err}} = p$.

4 Stage I: benchmarking the circuit-level noise

For a fixed circuit dimension ℓ , the number of failed tests obeys

$$Y_\ell \sim \text{Bin}(s_\ell, q_\ell). \quad (3)$$

The benchmark should output an upper confidence bound rather than only the raw frequency $\hat{q}_\ell$.

4.1 Hoeffding upper confidence bound

A simple one-sided bound is

$$q_\ell^{\text{U}} = \min \left\{ 1, \hat{q}_\ell + \sqrt{\frac{\log(1/\varepsilon_{\text{bench}})}{2s_\ell}} \right\}, \quad (4)$$

which satisfies

$$\Pr[q_\ell > q_\ell^{\text{U}}] \leq \varepsilon_{\text{bench}}. \quad (5)$$

4.2 Exact binomial upper confidence bound

For finite data, the script uses a one-sided Clopper–Pearson bound by default. For $0 \leq Y_\ell < s_\ell$, it is the $(1 - \varepsilon_{\text{bench}})$ quantile of

$$\text{Beta}(Y_\ell + 1, s_\ell - Y_\ell). \quad (6)$$

This bound is exact and generally tighter than Equation (4) for the data sizes relevant to benchmarking.

5 Stage II: converting test noise into computation error

From Equation (2), define

$$r_\ell^{\text{U}} = \min\{1, kq_\ell^{\text{U}}\}. \quad (7)$$

Under the worst-case effect model, the per-round computation error satisfies

$$p_\ell \leq p_\ell^{\text{U}} := c_\ell + (1 - c_\ell)r_\ell^{\text{U}} = c_\ell + (1 - c_\ell) \min\{1, kq_\ell^{\text{U}}\}. \quad (8)$$

The confidence statement is

$$\Pr[p_\ell > p_\ell^{\text{U}}] \leq \varepsilon_{\text{bench}}. \quad (9)$$

Remark 1. *If the experimental benchmark directly estimates the wrong-output probability of a computation round, use p_ℓ^{U} directly and omit the factor k . If it estimates the harmful-round probability r_ℓ , use $p_\ell^{\text{U}} = c_\ell + (1 - c_\ell)r_\ell^{\text{U}}$.*

6 Majority-vote correctness

Let d be odd and suppose every computation round is independently wrong with probability at most p_ℓ^{U} . The exact majority-vote error is

$$\varepsilon_{\text{vote}}^{\text{exact}}(d, p_\ell^{\text{U}}) = \sum_{j=(d+1)/2}^d \binom{d}{j} (p_\ell^{\text{U}})^j (1 - p_\ell^{\text{U}})^{d-j}. \quad (10)$$

For $p_\ell^{\text{U}} < 1/2$, Hoeffding’s inequality gives

$$\varepsilon_{\text{vote}}(d, p_\ell^{\text{U}}) \leq \exp\left[-2d \left(\frac{1}{2} - p_\ell^{\text{U}}\right)^2\right]. \quad (11)$$

The benchmark-confidence failure and the majority-vote failure are combined by a union bound:

$$\varepsilon_{\text{tot}} \leq \varepsilon_{\text{bench}} + \varepsilon_{\text{vote}}. \quad (12)$$

7 Performance-driven benchmark

Fix a target total correctness error $\varepsilon_\star$. Allocate

$$\varepsilon_{\text{vote}} = \varepsilon_\star - \varepsilon_{\text{bench}} > 0. \quad (13)$$

For every circuit dimension ℓ , the exact required number of computation rounds is

$$d_{\min}(\ell; \varepsilon_\star) = \min_{\substack{d \geq 1 \\ d \text{ odd}}} \left\{ d : \varepsilon_{\text{vote}}^{\text{exact}}(d, p_\ell^{\text{U}}) \leq \varepsilon_\star - \varepsilon_{\text{bench}} \right\}. \quad (14)$$

A closed-form sufficient choice follows from Equation (11):

$$d_{\min}^H(\ell; \varepsilon_\star) = \left\lceil \frac{\log(1/(\varepsilon_\star - \varepsilon_{\text{bench}}))}{2 \left(\frac{1}{2} - p_\ell^U\right)^2} \right\rceil_{\text{odd}}. \quad (15)$$

This is the performance-driven map

$$p_{\text{err}} \longmapsto d_{\min} \quad \text{at fixed } \varepsilon_\star. \quad (16)$$

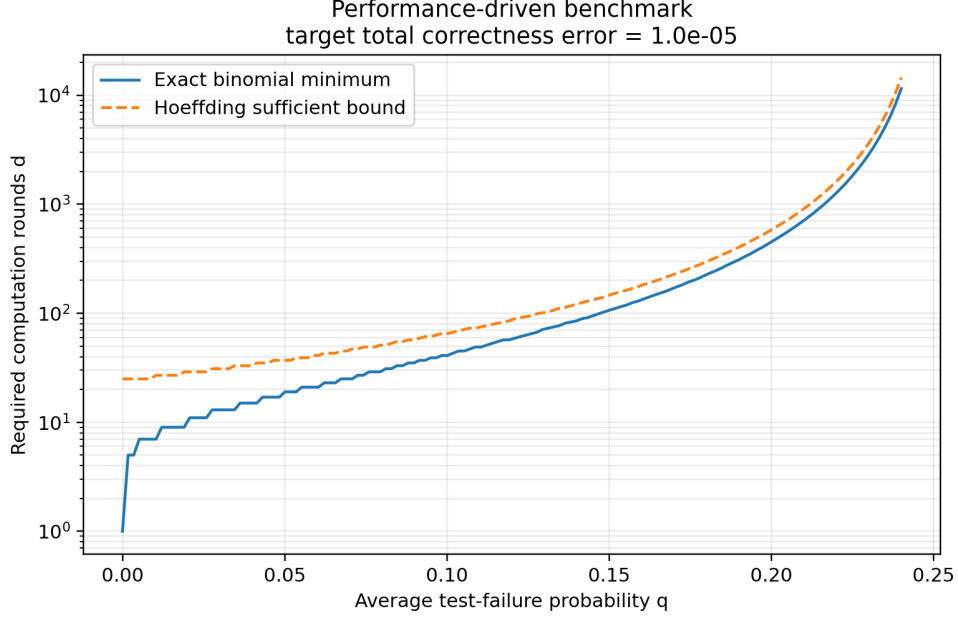


Figure 1: Performance-driven benchmark generated by the accompanying script for $k = 2$, $c = 0$, $\varepsilon_\star = 10^{-5}$, and $\varepsilon_{\text{bench}} = 10^{-6}$. The exact binomial calculation is compared with the sufficient Hoeffding bound.

8 Cost-driven benchmark

Fix an odd computation-round budget d_0 . For each circuit dimension ℓ , Equation (10) gives the exact achieved correctness error

$$\varepsilon_{\text{tot}}^{\text{exact}}(\ell; d_0) \leq \varepsilon_{\text{bench}} + \varepsilon_{\text{vote}}^{\text{exact}}(d_0, p_\ell^U). \quad (17)$$

The analytic upper bound is

$$\varepsilon_{\text{tot}}^H(\ell; d_0) \leq \varepsilon_{\text{bench}} + \exp \left[-2d_0 \left(\frac{1}{2} - p_\ell^U \right)^2 \right]. \quad (18)$$

This is the cost-driven map

$$p_{\text{err}} \longmapsto \varepsilon_{\text{tot}} \quad \text{at fixed } d_0. \quad (19)$$

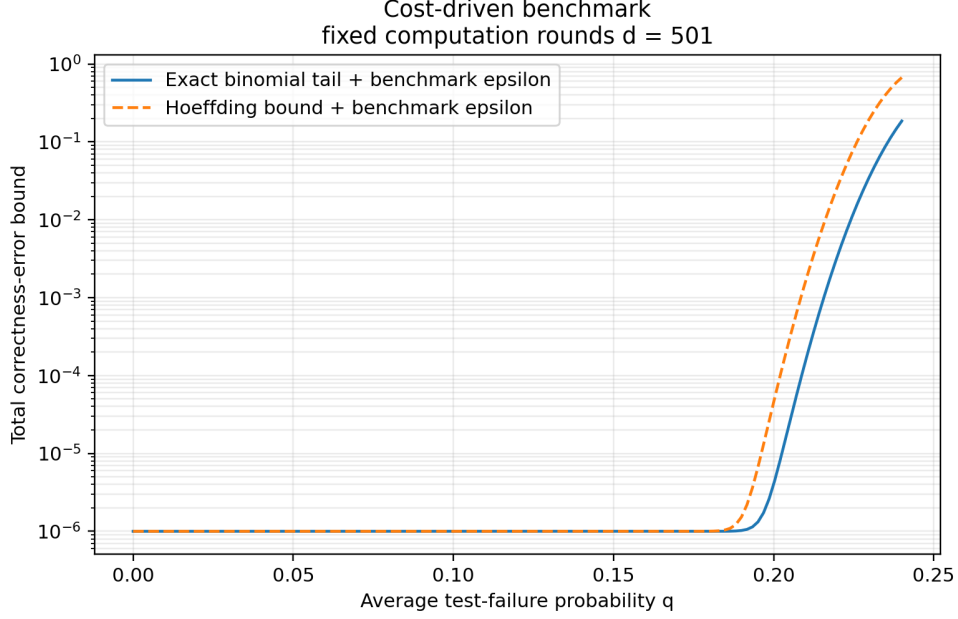


Figure 2: Cost-driven benchmark generated for a fixed budget $d_0 = 501$. The benchmark confidence error creates a floor at $\varepsilon_{\text{bench}} = 10^{-6}$.

9 Circuit-dimension maps and heatmaps

Circuit dimension does not enter the majority-vote equations directly. It enters through the experimentally measured noise:

$$\ell \mapsto q_\ell^U \mapsto p_\ell^U \mapsto \begin{cases} d_{\min}(\ell; \varepsilon_\star), & \text{performance-driven,} \\ \varepsilon_{\text{tot}}(\ell; d_0), & \text{cost-driven.} \end{cases} \quad (20)$$

For a scalar dimension ℓ , the script generates line plots. For circuits indexed by width and depth, it generates three heatmaps: the upper confidence bound on circuit-level noise, the required number of computation rounds at fixed $\varepsilon_\star$, and the achieved correctness error at fixed d_0 .

10 The 1/4 threshold for two test types

For $k = 2$ and $c = 0$, Equation (8) becomes

$$p_\ell^U = 2q_\ell^U. \quad (21)$$

Majority amplification requires $p_\ell^U < 1/2$, hence

$$q_\ell^U < \frac{1}{4}. \quad (22)$$

More generally, Equations (1) and (8) give

$$q_\ell^U < \frac{\alpha}{k} = \frac{1 - 2c_\ell}{k(2 - 2c_\ell)}. \quad (23)$$

The relevant quantity is the upper confidence bound q_ℓ^U , not only the observed frequency $\hat{q}_\ell$. As $q_\ell^U \uparrow 1/4$, the denominator in Equation (15) tends to zero and the required number of computation rounds diverges. Thus 1/4 is an asymptotic boundary in the conservative model, not a practical operating point.

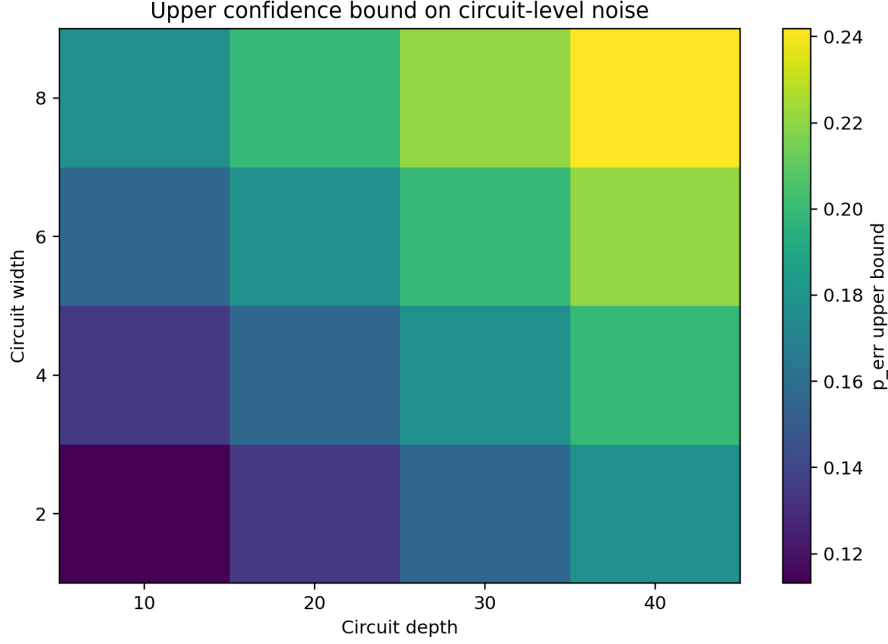


Figure 3: Example upper-confidence noise map over circuit width and depth. The example data are synthetic and are included only to demonstrate the workflow.

11 Connection with the structure of the proof

The reformulation uses three ingredients from the verification analysis:

1. **Correctness.** The paper bounds the probability that a majority of d honest computation rounds is wrong by $\exp[-2d(1/2 - c)^2]$. The benchmarked version replaces c by the data-dependent upper bound p_ℓ^U .
2. **Trap detection.** The security argument ensures that a harmful deviation is detected by at least one of the k trap types. This yields the conservative conversion $r_\ell \leq kq_\ell$.
3. **Robustness.** The paper treats noisy test failures through a concentration argument and requires the honest noise rate to lie below the accepted test threshold. In the i.i.d. benchmark, the threshold is replaced by a confidence interval inferred from a dedicated test block.

The malicious proof must retain a hidden random partition and maximize over correlated attacks. The present benchmark does neither: it estimates one fixed stationary parameter and extrapolates it to later computation rounds.

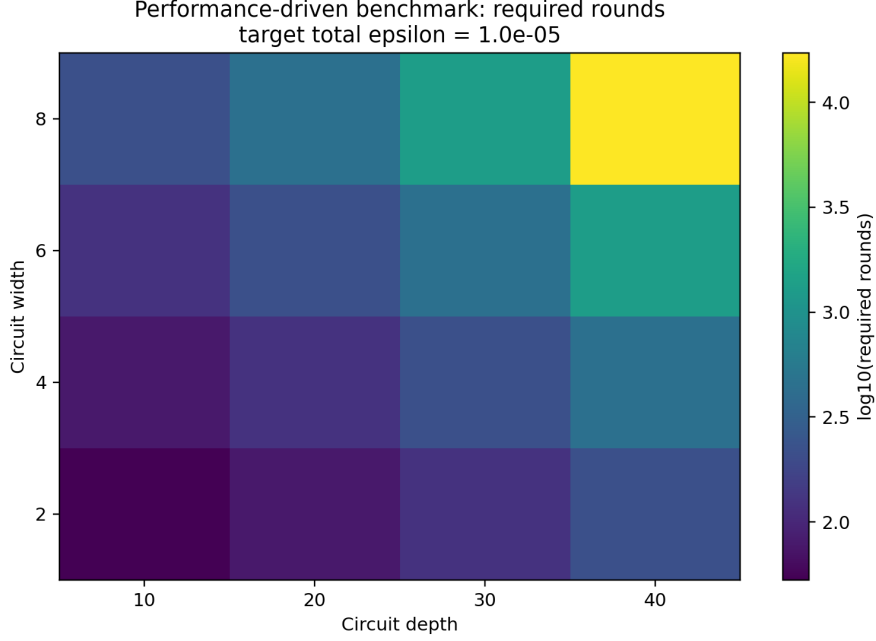


Figure 4: Performance-driven heatmap obtained by applying Equation (14) to every circuit instance. The color scale is logarithmic in the required number of rounds.

12 Recommended experimental workflow

Input: a circuit instance ℓ , number of tests s_ℓ , target confidence failure $\varepsilon_{\text{bench}}$, test count Y_ℓ , number of test types k , intrinsic ideal error c_ℓ , and either a target error $\varepsilon_\star$ or a fixed computation budget d_0 .

Step 1. Compute $\hat{q}_\ell = Y_\ell/s_\ell$ and a one-sided upper confidence bound q_ℓ^U .

Step 2. Convert it to $p_\ell^U = c_\ell + (1 - c_\ell) \min\{1, kq_\ell^U\}$, unless a tighter project-specific transfer function is available.

Step 3a, performance-driven. Find the smallest odd d satisfying Equation (14).

Step 3b, cost-driven. Evaluate Equation (17) at the available odd budget d_0 .

Output: for every circuit dimension, report $(\hat{q}_\ell, q_\ell^U, p_\ell^U, d_{\min}, \varepsilon_{\text{tot}}(d_0))$.

13 Limitations

The conclusions depend on stationarity, independence, and the test-to-computation transfer inequality. Correlated noise, drift between the benchmark block and the computation block, adaptive behavior, or circuit-dependent changes not captured by the selected dimension invalidate the simple binomial model. A practical implementation should therefore record time order, repeat benchmark blocks periodically, and test for drift or overdispersion before interpreting the resulting curves as correctness guarantees.

14 Software

The accompanying file `iid_benchmark_plots.py` implements:

- exact and Hoeffding majority-vote calculations;
- one-sided Clopper–Pearson and Hoeffding confidence bounds;

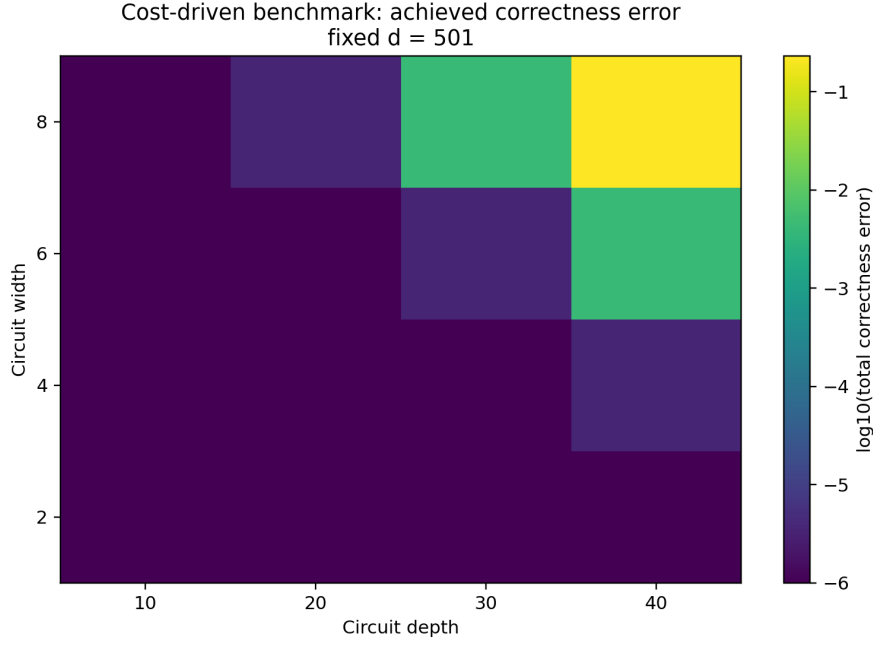


Figure 5: Cost-driven heatmap obtained by applying Equation (17) at fixed $d_0 = 501$. The color scale is logarithmic in the correctness-error bound.

- the three meanings of `p_err` described in Section 3;
- performance- and cost-driven plots;
- one-dimensional circuit-family plots;
- width/depth heatmaps and a derived CSV summary.

References

- [1] S. A. Sater and H. Ollivier. Composable Verification in the Circuit-Model via Magic-Blindness. arXiv:2601.07111v2, 2026.